pleasemendmy.com

Design Process Document

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1. Project Overview

1.1 Purpose

I have a lifelong habit of putting people on the spot, bringing up moral and philosophical questions in everyday conversation to test whether our assumptions hold up when examined. **pleasemendmy.com** is kind of an intimate project because it's an extension of that habit. I wanted to challenge you (yes, you 🫵) on how you use technology every day.

Today's hardware and browsers can hold a searchable encyclopedia of practical knowledge on the device and answer in milliseconds, so that tool should already exist and be the default. Instead, advertising incentives, platform politics, and pressure to route basic questions through AI products have left us with slow, bloated tools. As designers there should be nothing more heartbreaking than seeing our friends and family depend on tools that are slower, less economical, return AI answers that can change from one run to the next, and are increasingly becoming subscription services. Whatever happened to knowledge beeing free for all?

This project is meant to be kind of punk and countercultural. I built it to criticize slow, wasteful defaults by showing the alternative, and I've decided to maintain and improve it for future use!

1.2 Audience and Goals

The audience is anyone looking up practical repairs, household maintenance, or first aid.

1. **Fast answers:** Search runs in the browser, so a keystroke triggers no server request.
2. **Vetted sources:** Every guide comes from an existing human-written source (435,213 items). AI is used only offline to reformat text.
3. **Low waste:** The search index is stored on the device with `CacheStorage`, so a lookup needs no server computation.
4. **Two ways to ask:** A natural-language prompt ("*please, mend my [object]*") is paired with clickable part diagrams.

2. Research and Insights

2.1 From Calling Parents to Searching Online

A mentor conversation during this hackathon framed the problem. Young adults used to call their parents for help with things they could not find elsewhere, such as eating well, mending something broken, or caring for children. General advice was easy to find, but advice for a specific situation was hard.

Today people take the same questions to Google or AI. AI is good at specific, open-ended problems, but neither tool suits general knowledge lookup:

- **Search engines:** The user types a query, waits for results, opens a page, and scrolls to find the instructions.
- **AI chat:** The user waits while the model queries the web and writes a long answer. The answer is often not concise, and the user cannot easily tell whether it is accurate or based on outdated sources.

2.2 Vocabulary Gap

Users often do not know part names (a bicycle *derailleur*) and type symptoms in everyday words (*"it's not turning on"*). The prompt strips filler words while keeping negation, and the diagrams let a user point at a part instead of naming it.

3. Initial Mockups and Prototypes

1. **Category filters:** Buttons for browsing by category. These repeated the directory navigation the project is meant to avoid.
2. **Grid of 100 result cards:** Cards below the prompt. Scanning many cards before reading anything repeats the search-result workflow.
3. **Hero typography:** Sketches of the prompt as stitched thread, blueprint lines, and letters built from hammers, wrenches, and rulers, to give lookup a physical, hand-made look.

4. Iterations and Feedback

- **Inline answer:** The card grid was replaced by the top-ranked guide rendered under the prompt, with the next five results listed beside it as alternatives.
- **Input stability:** The fixed prompt text cannot be selected or edited, `Ctrl+A` selects only the user's text, and the previous answer stays visible until the next one is ready, so results do not move the caret.
- **No hardcoded lists:** Subject relationships, such as *Inspiron* to *laptop*, come from Open English WordNet across the whole catalog rather than a hand-picked list.
- **Offline reformatting:** A build-time pipeline asks a language model to restructure source HTML into tools, warnings, and steps under a strict schema. It ran on a 100-guide pilot, and each result is validated before it is published.

5. Final Product and Outcomes

- **Prompt:** A bare noun asks *"What's wrong with your [noun]?"* and shows its top guide immediately. A full symptom phrase searches directly.
- **Part diagrams:** Hovering, focusing, or selecting a part in the SVG or in the list highlights the other and searches that part.
- **Guides:** Each guide shows safety notices, tools, materials, steps, and a link to the original source.
- **On-device storage:** The catalog is kept in `CacheStorage` and checked against the server with an `ETag` request, which returns `304 Not Modified` when nothing has changed.

6. Figma

[TODO: paste the public Figma link here]

7. Citations and AI Use

7.1 Visuals

No AI was used to create visuals, UI graphics, typography, or diagrams. Hero wordplay, spotlight lighting, and layouts were made by hand. Diagrams come from Wikimedia Commons and Openclipart under open licenses, with source hashes and credits recorded.

7.2 Data Sources

- **Kiwix library:** ZIM archives of WikiMed (CC BY-SA 4.0), iFixit (CC BY-NC-SA 3.0), and Appropedia (CC BY-SA 3.0).
- **Other:** U.S. Army FM 4-25.11 First Aid (public domain), the wikiHow corpus (CC BY-NC-SA 3.0), and Open English WordNet (CC BY 4.0).

7.3 AI Use

- **Code:** Claude and Gemini helped write code and tests.
- **Offline text:** `qwen/qwen3.6-35b-a3b` reformatted the 100-guide pilot into structured JSON, and was used in offline experiments that propose which shape in an existing diagram matches a part name. No model runs when a user searches.